ADITYA MAHAMUNI

APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

Applied AI Engineer with experience designing reliable AI systems, developing scalable Python services, and translating emerging machine learning capabilities into practical software solutions. Combines strong software engineering fundamentals with expertise in agentic AI, LLM applications, evaluation, and cloud-based deployment; experienced in cross-functional engineering and delivering solutions in complex environments.

TECHNICAL EXPERTISE

Applied AI: Agentic AI, LLM Applications, Tool Use, RAG, Retrieval, Context/Prompt Engineering, Structured Outputs, MCP, Orchestration; LangGraph, OpenAI Agents SDK

Evaluation & ML: AI & LLM Evaluation, Synthetic Data Generation, NLP, Transformers, PyTorch, TensorFlow, Fine-Tuning and LoRA

Engineering & Platforms: Python, C++, SQL, FastAPI, Pydantic, PostgreSQL, Kafka, REST, JWT/RBAC, Docker, Linux, GitLab CI, Jenkins, AWS Bedrock / SageMaker, Git

EXPERIENCE

NatWest, London, UK | **Machine Learning Engineer** · *Dec 2024 - Present*

- Built a synthetic-data pipeline for complex, multi-system workflows, enabling rapid evaluation experiments, model regression analysis, and adversarial testing without production data; reduced issue diagnosis and resolution turnaround time by 70%.
- Developed a production FastAPI service for automated complaints investigation using AI agents, integrating event-driven orchestration, execution tracking, persistence, and feedback APIs.
- Built workflows using observability tools and evaluation frameworks to assess LLM and agent outputs, retrieval, tool use, and failure modes, supporting reliable iteration and regression investigation.
- Prototyped LangGraph, OpenAI Agents SDK, smolagents, and AutoGen; designed reusable routing, handoff, parallel, and evaluator-optimizer patterns and assessed production trade-offs.
- Built MCP tools and integrations for knowledge retrieval, developer platforms, and custom services to support multi-step agent workflows.
- Led cross-team delivery, owned technical design decisions, and communicated progress and outcomes to stakeholders and leadership.
- Fine-tuned a language model using LoRA for COBOL code completion, infilling, and domain-specific question answering; evaluated task performance and integrated inference through AWS SageMaker endpoints into agent-assisted developer workflows.

NVIDIA Corporation, Pune, India | **Software Developer** · *Jul 2019 - Aug 2023*

- Developed a scalable Python framework and training scenarios for autonomous-vehicle simulation to support scenario-driven development and testing.
- Optimised Python, GitLab CI, and Docker delivery pipelines to improve deployment efficiency and accelerate scenario updates.
- Built Jenkins/Python automated testing frameworks that reduced manual testing effort and shortened test cycles.
- Developed ML tooling and CI/CD workflows using NLP, PyTorch, and TensorFlow to support development and experimentation.
- Created a Python/Linux diagnostic tool that reduced platform-specific issues and improved debugging reliability.

PROJECTS

Financial Event Extraction Using NLP | MSc Dissertation

Applied zero-shot and few-shot LLM approaches and fine-tuning to extract and classify financial events from news articles; compared generative modelling approaches for a financial-language task.

Financial Health and Stock Performance Analysis

Built Python-based financial analysis using clustering and regression to explore financial health and market performance. Original project reported 99% classification accuracy and regression RMSE of 1.118; task-specific methodology available in project materials.

Protein Expression Prediction

Developed image-based feature extraction using PCA, GLCM, and transfer learning; experimented with OLS, SVR, and CNN approaches to predict tissue protein expression, reporting CNN RMSE of 0.677.

EDUCATION

University of Warwick, UK | MSc Computer Science (2:1) | Sep 2023 - Sep 2024

Government College of Engineering, Aurangabad, India | BE Information Technology (7.448/10) | Jul 2015 - May 2019

CERTIFICATION

Neural Networks and Deep Learning - DeepLearning.AI | Sep 2020